Example 1.8

The mean breaking strength of cables supplied by a manufacturer is 1900 n/m^2 with a standard deviation of 120 n/m^2. The manufacturer introduced a new technique in the manufacturing process and claimed that the breaking strength of the cables has increased. In order to test the claim, a sample of 60 cables is tested. It is found that the mean breaking strength of the sampled cables is 1960 n/m^2. Can we support the claim at 1% level of significance?

Solution:

Step 1 : Let μ and σ represent respectively the mean and standard deviation of the probability distribution of the breaking strength of the cables. It is given that $\sigma = 120$ n/m^2. The null and alternative hypotheses are

Null hypothesis H_0: $\mu = 1900$

i.e., the mean breaking strength of the cables is not significantly different from 1900n/m^2.

Alternative hypothesis: H_1: $\mu > 1900$

i.e., the mean breaking strength of the cables is significantly more than 1900n/m^2.

It may be noted that it is a one-sided (right) alternative hypothesis.

Step 2 : **Data**

The given sample information are

Sample size (n) = 60. Hence, it is a large sample.

Sample mean ($\bar{x}$)= 1960

Step 3 : **Level of significance**

$\alpha = 1\%$

Step 4 : **Test statistic**

The test statistic is $Z = \dfrac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$, under H_0

Since n is large, under the null hypothesis, the sampling distribution of Z is the $N(0,1)$ distribution.

Step 5 : **Calculation of test statistic**

The value of Z under H_0 is calculated from $z_0 = \dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$

$$z_0 = \frac{1960 - 1900}{120 / \sqrt{60}}$$

Thus, $z_0 = 3.87$

Step 6 : **Critical value**

Since H_1 is a one-sided (right) alternative hypothesis, the critical value at $\alpha = 0.01$ level of significance is $z_e = z_{0.01} = 2.33$ (*see* Table 1.6)

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